

9.2 RUNGE-KUTTA METHODS

REVIEW MATERIAL

- Section 2.8 (see page 78)

INTRODUCTION Probably one of the more popular as well as most accurate numerical procedures used in obtaining approximate solutions to a first-order initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$ is the **fourth-order Runge-Kutta method**. As the name suggests, there are Runge-Kutta methods of different orders.

RUNGE-KUTTA METHODS Fundamentally, all Runge-Kutta methods are generalizations of the basic Euler formula (1) of Section 9.1 in that the slope function f is replaced by a weighted average of slopes over the interval $x_n \leq x \leq x_{n+1}$. That is,

$$y_{n+1} = y_n + h \overbrace{(w_1 k_1 + w_2 k_2 + \cdots + w_m k_m)}^{\text{weighted average}}. \quad (1)$$

Here the weights w_i , $i = 1, 2, \dots, m$, are constants that generally satisfy $w_1 + w_2 + \cdots + w_m = 1$, and each k_i , $i = 1, 2, \dots, m$, is the function f evaluated at a selected point (x, y) for which $x_n \leq x \leq x_{n+1}$. We shall see that the k_i are defined recursively. The number m is called the **order** of the method. Observe that by taking $m = 1$, $w_1 = 1$, and $k_1 = f(x_n, y_n)$, we get the familiar Euler formula $y_{n+1} = y_n + hf(x_n, y_n)$. Hence Euler's method is said to be a **first-order Runge-Kutta method**.

The average in (1) is not formed willy-nilly, but parameters are chosen so that (1) agrees with a Taylor polynomial of degree m . As we saw in the preceding section, if a function $y(x)$ possesses $k + 1$ derivatives that are continuous on an open interval containing a and x , then we can write

$$y(x) = y(a) + y'(a) \frac{x-a}{1!} + y''(a) \frac{(x-a)^2}{2!} + \cdots + y^{(k+1)}(c) \frac{(x-a)^{k+1}}{(k+1)!},$$

where c is some number between a and x . If we replace a by x_n and x by $x_{n+1} = x_n + h$, then the foregoing formula becomes

$$y(x_{n+1}) = y(x_n + h) = y(x_n) + hy'(x_n) + \frac{h^2}{2!} y''(x_n) + \cdots + \frac{h^{k+1}}{(k+1)!} y^{(k+1)}(c),$$

where c is now some number between x_n and x_{n+1} . When $y(x)$ is a solution of $y' = f(x, y)$ in the case $k = 1$ and the remainder $\frac{1}{2} h^2 y''(c)$ is small, we see that a Taylor polynomial $y(x_{n+1}) = y(x_n) + hy'(x_n)$ of degree one agrees with the approximation formula of Euler's method

$$y_{n+1} = y_n + hy'_n = y_n + hf(x_n, y_n).$$

A SECOND-ORDER RUNGE-KUTTA METHOD To further illustrate (1), we consider now a **second-order Runge-Kutta procedure**. This consists of finding constants or parameters w_1 , w_2 , α , and β so that the formula

$$y_{n+1} = y_n + h(w_1 k_1 + w_2 k_2), \quad (2)$$

where

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \alpha h, y_n + \beta h k_1),$$

agrees with a Taylor polynomial of degree two. For our purposes it suffices to say that this can be done whenever the constants satisfy

$$w_1 + w_2 = 1, \quad w_2 \alpha = \frac{1}{2}, \quad \text{and} \quad w_2 \beta = \frac{1}{2}. \quad (3)$$

This is an algebraic system of three equations in four unknowns and has infinitely many solutions:

$$w_1 = 1 - w_2, \quad \alpha = \frac{1}{2w_2}, \quad \text{and} \quad \beta = \frac{1}{2w_2}, \quad (4)$$

where $w_2 \neq 0$. For example, the choice $w_2 = \frac{1}{2}$ yields $w_1 = \frac{1}{2}$, $\alpha = 1$, and $\beta = 1$, and so (2) becomes

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2),$$

where $k_1 = f(x_n, y_n)$ and $k_2 = f(x_n + h, y_n + hk_1)$.

Since $x_n + h = x_{n+1}$ and $y_n + hk_1 = y_n + hf(x_n, y_n)$, the foregoing result is recognized to be the improved Euler's method that is summarized in (3) and (4) of Section 9.1.

In view of the fact that $w_2 \neq 0$ can be chosen arbitrarily in (4), there are many possible second-order Runge-Kutta methods. See Problem 2 in Exercises 9.2.

We shall skip any discussion of third-order methods in order to come to the principal point of discussion in this section.

A FOURTH-ORDER RUNGE-KUTTA METHOD A fourth-order Runge-Kutta procedure consists of finding parameters so that the formula

$$y_{n+1} = y_n + h(w_1 k_1 + w_2 k_2 + w_3 k_3 + w_4 k_4), \quad (5)$$

where

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \alpha_1 h, y_n + \beta_1 h k_1)$$

$$k_3 = f(x_n + \alpha_2 h, y_n + \beta_2 h k_1 + \beta_3 h k_2)$$

$$k_4 = f(x_n + \alpha_3 h, y_n + \beta_4 h k_1 + \beta_5 h k_2 + \beta_6 h k_3),$$

agrees with a Taylor polynomial of degree four. This results in a system of 11 equations in 13 unknowns. The most commonly used set of values for the parameters yields the following result:

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), \\ k_1 &= f(x_n, y_n) \\ k_2 &= f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}h k_1\right) \\ k_3 &= f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}h k_2\right) \\ k_4 &= f(x_n + h, y_n + h k_3). \end{aligned} \quad (6)$$

While other fourth-order formulas are easily derived, the algorithm summarized in (6) is so widely used and recognized as a valuable computational tool it is often referred to as *the* fourth-order Runge-Kutta method or *the classical* Runge-Kutta method. It is (6) that we have in mind, hereafter, when we use the abbreviation *the RK4 method*.

You are advised to look carefully at the formulas in (6); note that k_2 depends on k_1 , k_3 depends on k_2 , and k_4 depends on k_3 . Also, k_2 and k_3 involve approximations to the slope at the midpoint $x_n + \frac{1}{2}h$ of the interval defined by $x_n \leq x \leq x_{n+1}$.

EXAMPLE 1 RK4 Method

Use the RK4 method with $h = 0.1$ to obtain an approximation to $y(1.5)$ for the solution of $y' = 2xy$, $y(1) = 1$.

SOLUTION For the sake of illustration let us compute the case when $n = 0$. From (6) we find

$$\begin{aligned} k_1 &= f(x_0, y_0) = 2x_0y_0 = 2 \\ k_2 &= f\left(x_0 + \frac{1}{2}(0.1), y_0 + \frac{1}{2}(0.1)2\right) \\ &= 2\left(x_0 + \frac{1}{2}(0.1)\right)\left(y_0 + \frac{1}{2}(0.2)\right) = 2.31 \\ k_3 &= f\left(x_0 + \frac{1}{2}(0.1), y_0 + \frac{1}{2}(0.1)2.31\right) \\ &= 2\left(x_0 + \frac{1}{2}(0.1)\right)\left(y_0 + \frac{1}{2}(0.231)\right) = 2.34255 \\ k_4 &= f(x_0 + (0.1), y_0 + (0.1)2.34255) \\ &= 2(x_0 + 0.1)(y_0 + 0.234255) = 2.715361 \end{aligned}$$

TABLE 9.5 RK4 Method with $h = 0.1$

x_n	y_n	Actual value	Abs. error	% Rel. error
1.00	1.0000	1.0000	0.0000	0.00
1.10	1.2337	1.2337	0.0000	0.00
1.20	1.5527	1.5527	0.0000	0.00
1.30	1.9937	1.9937	0.0000	0.00
1.40	2.6116	2.6117	0.0001	0.00
1.50	3.4902	3.4904	0.0001	0.00

and therefore

$$\begin{aligned} y_1 &= y_0 + \frac{0.1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\ &= 1 + \frac{0.1}{6}(2 + 2(2.31) + 2(2.34255) + 2.715361) = 1.23367435. \end{aligned}$$

The remaining calculations are summarized in Table 9.5, whose entries are rounded to four decimal places. ■

Inspection of Table 9.5 shows why the fourth-order Runge-Kutta method is so popular. If four-decimal-place accuracy is all that we desire, there is no need to use a smaller step size. Table 9.6 compares the results of applying Euler's, the improved Euler's, and the fourth-order Runge-Kutta methods to the initial-value problem $y' = 2xy$, $y(1) = 1$. (See Tables 9.1 and 9.3.)

TABLE 9.6 $y' = 2xy$, $y(1) = 1$

Comparison of numerical methods with $h = 0.1$					Comparison of numerical methods with $h = 0.05$				
x_n	Euler	Improved Euler	RK4	Actual value	x_n	Euler	Improved Euler	RK4	Actual value
1.00	1.0000	1.0000	1.0000	1.0000	1.00	1.0000	1.0000	1.0000	1.0000
1.10	1.2000	1.2320	1.2337	1.2337	1.05	1.1000	1.1077	1.1079	1.1079
1.20	1.4640	1.5479	1.5527	1.5527	1.10	1.2155	1.2332	1.2337	1.2337
1.30	1.8154	1.9832	1.9937	1.9937	1.15	1.3492	1.3798	1.3806	1.3806
1.40	2.2874	2.5908	2.6116	2.6117	1.20	1.5044	1.5514	1.5527	1.5527
1.50	2.9278	3.4509	3.4902	3.4904	1.25	1.6849	1.7531	1.7551	1.7551
					1.30	1.8955	1.9909	1.9937	1.9937
					1.35	2.1419	2.2721	2.2762	2.2762
					1.40	2.4311	2.6060	2.6117	2.6117
					1.45	2.7714	3.0038	3.0117	3.0117
					1.50	3.1733	3.4795	3.4903	3.4904

TRUNCATION ERRORS FOR THE RK4 METHOD In Section 9.1 we saw that global truncation errors for Euler's method and for the improved Euler's method are, respectively, $O(h)$ and $O(h^2)$. Because the first equation in (6) agrees with a Taylor polynomial of degree four, the local truncation error for this method is $y^{(5)}(c) h^5/5!$ or $O(h^5)$, and the global truncation error is thus $O(h^4)$. It is now obvious why Euler's method, the improved Euler's method, and (6) are *first*-, *second*-, and *fourth*-order Runge-Kutta methods, respectively.

EXAMPLE 2 Bound for Local Truncation Errors

Find a bound for the local truncation errors for the RK4 method applied to $y' = 2xy$, $y(1) = 1$.

SOLUTION By computing the fifth derivative of the known solution $y(x) = e^{x^2-1}$, we get

$$y^{(5)}(c) \frac{h^5}{5!} = (120c + 160c^3 + 32c^5)e^{c^2-1} \frac{h^5}{5!}. \quad (7)$$

Thus with $c = 1.5$, (7) yields a bound of 0.00028 on the local truncation error for each of the five steps when $h = 0.1$. Note that in Table 9.5 the error in y_1 is much less than this bound.

Table 9.7 gives the approximations to the solution of the initial-value problem at $x = 1.5$ that are obtained from the RK4 method. By computing the value of the analytic solution at $x = 1.5$, we can find the error in these approximations. Because the method is so accurate, many decimal places must be used in the numerical solution to see the effect of halving the step size. Note that when h is halved, from $h = 0.1$ to $h = 0.05$, the error is divided by a factor of about $2^4 = 16$, as expected. ■

TABLE 9.7 RK4 Method

h	Approx.	Error
0.1	3.49021064	$1.32321089 \times 10^{-4}$
0.05	3.49033382	$9.13776090 \times 10^{-6}$

ADAPTIVE METHODS We have seen that the accuracy of a numerical method for approximating solutions of differential equations can be improved by decreasing the step size h . Of course, this enhanced accuracy is usually obtained at a cost—namely, increased computation time and greater possibility of round-off error. In general, over the interval of approximation there may be subintervals where a relatively large step size suffices and other subintervals where a smaller step is necessary to keep the truncation error within a desired limit. Numerical methods that use a variable step size are called **adaptive methods**. One of the more popular of the adaptive routines is the **Runge-Kutta-Fehlberg method**. Because Fehlberg employed two Runge-Kutta methods of differing orders, a fourth- and a fifth-order method, this algorithm is frequently denoted as the **RKF45 method**.*

*The Runge-Kutta method of order four used in RKF45 is *not* the same as that given in (6).

EXERCISES 9.2

Answers to selected odd-numbered problems begin on page ANS-15.

- Use the RK4 method with $h = 0.1$ to approximate $y(0.5)$, where $y(x)$ is the solution of the initial-value problem $y' = (x + y - 1)^2$, $y(0) = 2$. Compare this approximate value with the actual value obtained in Problem 11 in Exercises 9.1.
- Assume that $w_2 = \frac{3}{4}$ in (4). Use the resulting second-order Runge-Kutta method to approximate $y(0.5)$, where $y(x)$ is the solution of the initial-value problem in Problem 1. Compare this approximate value with the approximate value obtained in Problem 11 in Exercises 9.1.

In Problems 3–12 use the RK4 method with $h = 0.1$ to obtain a four-decimal approximation of the indicated value.

- $y' = 2x - 3y + 1$, $y(1) = 5$; $y(1.5)$
- $y' = 4x - 2y$, $y(0) = 2$; $y(0.5)$
- $y' = 1 + y^2$, $y(0) = 0$; $y(0.5)$

- $y' = x^2 + y^2$, $y(0) = 1$; $y(0.5)$
- $y' = e^{-y}$, $y(0) = 0$; $y(0.5)$
- $y' = x + y^2$, $y(0) = 0$; $y(0.5)$
- $y' = (x - y)^2$, $y(0) = 0.5$; $y(0.5)$
- $y' = xy + \sqrt{y}$, $y(0) = 1$; $y(0.5)$
- $y' = xy^2 - \frac{y}{x}$, $y(1) = 1$; $y(1.5)$
- $y' = y - y^2$, $y(0) = 0.5$; $y(0.5)$

- If air resistance is proportional to the square of the instantaneous velocity, then the velocity v of a mass m dropped from a given height is determined from

$$m \frac{dv}{dt} = mg - kv^2, \quad k > 0.$$

Let $v(0) = 0$, $k = 0.125$, $m = 5$ slugs, and $g = 32$ ft/s².